Highlights

A multi-period vehicle routing problem with dynamic node-road damage for emergency material delivery

- MP-DDVRP models staged vehicle dispatch, period inventory, and dynamic node-road damage.
- IHGA adds damage-critical and period-aware repair to a dynamic-damage HGA framework.
- Paired tests indicate lower mean dynamic damage than VNS and IMA on representative C/R/RC cases.
- Ablation identifies damage-critical repair as the main algorithmic improvement over Base HGA.
- Sensitivity results quantify optimistic aggregation bias and early-resource dilution in multi-period dispatch.

A multi-period vehicle routing problem with dynamic node-road damage for emergency material delivery

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Abstract

Emergency material delivery after an earthquake is affected not only by routing distance, but also by time-dependent damage at demand nodes, damaged roads, and the timing at which rescue resources become available. This paper extends dynamic damage routing to a multi-period setting in which vehicles and relief inventories are released in dispatch waves rather than being fully available at time zero. The resulting multi-period dynamic damage vehicle routing problem (MP-DDVRP) couples route sequencing, road selection, period assignment, and cumulative period-inventory feasibility. To solve the problem, we develop an improved hybrid genetic algorithm (IHGA) that retains the dynamic node-road damage evaluation framework and adds damage-critical and period-aware repair moves for staged dispatch. On representative C, R, and RC instances, IHGA achieves lower mean dynamic damage than ESWA-IMA and Omega-VNS in the tested paired runs, with average reductions of 1.19% and 6.69%, respectively. The ablation results indicate that damage-critical repair provides the main improvement over the base HGA, while period-aware and route-structure moves mainly support feasibility and robustness in staged dispatch. Sensitivity experiments show that a single-period aggregate model gives an optimistic lower-bound estimate because it assumes that all vehicles and supplies can be used immediately. Under fixed total resources, staged multi-period release increases dynamic damage when early-period vehicle and inventory availability is diluted, highlighting the importance of modeling resource timing in post-earthquake routing decisions.

Keywords: Multi-period vehicle routing, emergency logistics, dynamic damage, hybrid genetic algorithm, sensitivity analysis, Solomon instances

1. Introduction

Emergency material distribution differs from ordinary vehicle routing because delay has a direct loss consequence. After a disaster, affected locations continuously change their states: a demand point may deteriorate if it is not served, while the arrival of supplies may reduce the expected loss. Transportation links also change over time. A damaged road can reduce travel speed and delay all nodes scheduled after that link. Therefore, minimizing geographical distance alone is insufficient for disaster-response routing. The routing policy must decide not only which vehicle serves which demand node, but also in which period the node is served, which link alternative is selected, and how limited supplies are allocated over time.

The multi-period structure is not a secondary parameter in this setting. In realistic emergency response, all vehicles and supplies are rarely available at time zero. Rescue capacity is released in waves because vehicles return, roads are cleared gradually, and inventories arrive through staged procurement or upstream transport. If the planning horizon is collapsed into a single period, the model implicitly allows later-period vehicles and supplies to be used immediately, creating an optimistic dispatch plan and hiding the loss caused by delayed resource availability. A multi-period model is therefore needed to represent when resources can actually be used, not only how many resources exist in total.

This study considers a multi-period vehicle routing problem with dynamic damage. The problem is motivated by emergency material delivery where a dispatch center repeatedly sends rescue vehicles to affected nodes. Each node has demand, service time, initial damage, deterioration rate, and recovery rate. Each pair of nodes is connected by multiple road alternatives, and each road alternative has an initial damage and deterioration trajectory. The resulting decision problem couples vehicle routing, period assignment, road selection, period inventory allocation, and dynamic loss evaluation.

The main challenge is that the objective is not a static sum of arc distances. The cost of serving a node depends on its arrival time, and the arrival time itself depends on previous node choices, service times, and road damage. This feedback creates a highly nonlinear combinatorial problem. Moreover, sensitivity analysis is important for emergency planning: decision makers often need to know how much benefit is obtained by adding vehicles, increasing the number of periods, or changing available inventory. A good algorithm should therefore deliver not only low objective values but also stable and interpretable marginal-effect curves.

The contributions of this paper are summarized as follows.

- We formulate MP-DDVRP, a multi-period extension of dynamic damage routing in which staged vehicle dispatch, period inventory, dynamic node damage, and dynamic road damage jointly determine rescue loss.
- We design IHGA for MP-DDVRP by combining route-based genetic search with damage-critical repair, period-aware relocation, and route-structure adjustment under cumulative inventory feasibility.

- We evaluate the model and algorithm through algorithm comparison, operator ablation, objective-proxy comparison, multi-period resource sensitivity, and node-priority analysis on representative and external C/R/RC instances.

Table 1 summarizes the relationship between this study and the original DCCVRPMRS model of Zhao et al. (2026). The present work inherits the dynamic node-road damage and multi-road selection setting, but changes the resource-availability assumption from a single dispatch horizon to staged multi-period release.

Table 1: Relationship with the original DCCVRPMRS study.

Aspect	Original DCCVRPMRS (Zhao et al., 2026)	This study
Damage representation	Dynamic node and road damage with multi-road selection.	Retains dynamic node-road damage and multi-road selection.
Resource timing	Vehicles and supplies are treated within a single dispatch horizon.	Vehicles and inventories are released over multiple dispatch periods.
Key feasibility condition	Vehicle capacity and route feasibility.	Vehicle capacity plus cumulative period-inventory feasibility.
Algorithmic focus	Route-similarity crossover and route/road local search.	Damage-critical and period-aware repair under staged dispatch.
Decision insight	Priority among damaged nodes and road alternatives.	Bias of single-period aggregation and value of modeling resource timing.

2. Literature review

The vehicle routing problem (VRP) and its variants have been extensively studied since the early development of vehicle routing with time windows (Solomon, 1987). Classical VRP models usually optimize distance, travel time, or fleet size under vehicle-capacity and service constraints. Surveys and benchmark studies show that high-quality solutions typically require hybrid metaheuristics that combine population-based search with local improvement procedures (Bräysy and Gendreau, 2005; Pisinger and Ropke, 2007; Vidal et al., 2013).

Emergency logistics introduces additional modeling requirements. Relief distribution after disasters is commonly dynamic and time dependent because demand, vehicle availability, and network accessibility change during the response process (Özdamar et al., 2004; Sheu, 2007). Literature reviews also show that post-disaster logistics models frequently combine routing, allocation, inventory, and transportation decisions rather than treating delivery as a static distance-minimization problem (Caunhye et al., 2012; Afshar and Haghani, 2012).

A second important stream concerns the objective function of humanitarian logistics. Since delayed delivery can translate into human suffering, response-time proxies, unmet-demand penalties, deprivation costs, and social-cost objectives have been proposed to better represent disaster-response performance (Holguín-Veras et al., 2013). In such settings, the timing of service is central: serving the same node earlier or later may cause very different losses.

This paper is positioned at the intersection of multi-period routing and time-dependent damage evaluation. Compared with distance-minimizing VRP models, the proposed problem evaluates the dynamic state of demand nodes and road links. Compared with standard time-window routing, the loss is a continuous function of arrival time instead of a hard feasibility interval. Compared with single-period emergency routing, the model explicitly represents repeated dispatch periods and period-level inventory constraints, which prevents later resources from being treated as if they were available in the first dispatch wave.

3. Problem description and modeling

Let $G = (N_0, A)$ be a complete directed graph, where $N = \{1, \ldots, n\}$ is the set of affected demand nodes and node 0 denotes the depot. The planning horizon is divided into P dispatch periods. Period p starts at $T_p = (p-1)\Delta$, where Δ is the period-length parameter. In the reported experiments, $\Delta = 1440/P$, corresponding to a normalized one-day planning horizon measured in minutes. In each period $p \in \{1, \ldots, P\}$, at most K rescue vehicles can be used. Each vehicle has capacity Q, and the available period inventory is B_p. In the implementation used in the experiments, B_p is set adaptively according to total demand and the number of periods.

The solution determines the assignment of every demand node to exactly one vehicle route, the service sequence within each route, the dispatch period of each route, and the road alternative selected for each traversed arc. Demand splitting is not allowed: once a node is assigned to a route, the vehicle delivers the full demand of that node. We refer to this staged dynamic-damage routing extension as MP-DDVRP.

Each demand node $i \in N$ has demand q_i, service time s_i, initial damage a_i, deterioration rate b_i, and recovery rate ρ_i. If node i is served at arrival time A_i, its remaining damage is evaluated as

$$D_i(A_i) = a_i e^{-\rho_i A_i} + \frac{b_i}{\rho_i}\left(1 - e^{-\rho_i A_i}\right). \tag{1}$$

This expression captures two competing effects: existing damage is gradually recovered, while unserved demand continues to deteriorate until service.

For each arc $(i, j) \in A$, several road alternatives $z \in Z$ are available. Road alternative z has initial damage a^r_{ijz}, deterioration rate b^r_{ijz}, and recovery rate ρ^r_{ij}. If a vehicle enters the road at time t, the road damage is

$$R_{ijz}(t) = a^r_{ijz} e^{-\rho^r_{ij} t} + \frac{b^r_{ijz}}{\rho^r_{ij}}\left(1 - e^{-\rho^r_{ij} t}\right). \tag{2}$$

The road state affects the effective speed:

$$v_{ijz}(t) = \left(1 - \frac{R_{ijz}(t)\rho^r_{\min}}{0.814\, b^r_{\max}}\right) v_0, \tag{3}$$

where v_0 is the nominal vehicle speed, $\rho^r_{\min}$ is the minimum road recovery rate, and $b^r_{\max}$ is the maximum road deterioration rate.

Consider a route $r = (0, i_1, i_2, \ldots, i_m)$ dispatched in period p. Let T_p be the start time offset of period p. The arrival time is recursively computed by

$$A_{i_1} = T_p + \frac{d_{0i_1z}}{v_{0i_1z}(T_p)}, \tag{4}$$

$$A_{i_{h+1}} = A_{i_h} + s_{i_h} + \frac{d_{i_h i_{h+1} z}}{v_{i_h i_{h+1} z}(A_{i_h} + s_{i_h})}, \quad h = 1, \ldots, m-1. \tag{5}$$

The primary objective is to minimize total weighted node damage:

$$\min F = \sum_{i \in N} \omega_i D_i(A_i) + M \cdot \Phi, \tag{6}$$

where ω_i is the loss weight of node i, M is a large penalty coefficient, and Φ is the total violation of vehicle-capacity and cumulative period-inventory constraints. The reported experiments use $M = 10000$. Road damage is not ignored: it changes speed through Eq. (3), thereby affecting arrival times and downstream node losses.

The main constraints are as follows:

$$\sum_{r} \sum_{h:i_h = i} 1 = 1, \qquad \forall i \in N, \tag{7}$$

$$\sum_{i \in r} q_i \le Q, \qquad \forall r, \tag{8}$$

$$\sum_{\tau=1}^{p} \sum_{r \in \mathcal{R}_\tau} \sum_{i \in r} q_i \le \sum_{\tau=1}^{p} B_\tau, \qquad p = 1, \ldots, P, \tag{9}$$

$$|\mathcal{R}_p| \le K, \qquad p = 1, \ldots, P. \tag{10}$$

The cumulative inventory constraint allows early periods to use only the supplies that have become available by that time. This is the key distinction between the proposed multi-period formulation and an aggregated single-period formulation: the latter only checks whether total supply is sufficient over the whole horizon, whereas the former also checks whether enough supply is available before each dispatch wave. As a result, the model can quantify the loss caused by delayed resource release.

4. Improved hybrid genetic algorithm

The proposed IHGA uses a route-based encoding. An individual contains the complete set of routes over all periods. Route index 1 to K belongs to period 1, route index $K + 1$ to $2K$ belongs to period 2, and so on. This fixed route-period mapping is important because route collapse or route renumbering would destroy the multi-period semantics.

4.1. Initialization

The initial candidate solutions are generated by combining randomized construction, urgency-aware construction, and distance-structured construction. The urgency score of node i is

$$u_i = a_i + \frac{b_i}{\max(\rho_i, \epsilon)}, \tag{11}$$

where ϵ avoids division by zero. Candidate populations combine randomized sequences, weighted urgency sequences, and urgency-ranked sequences with band-level shuffling. During insertion, candidate vehicles are screened by vehicle capacity and cumulative period-inventory feasibility. The insertion cost combines travel distance, current route load, and current route length.

4.2. Crossover and local search

The crossover operator identifies two similar routes from two parent solutions and preserves their common nodes. Nodes not preserved are reinserted into feasible routes. This route-similarity crossover is designed to inherit useful route blocks without breaking feasibility.

Local search is applied after crossover. The neighborhoods include intra-route swap, insertion, 2-opt reversal, inter-route exchange, and inter-route relocation. These operators are guided by initial damage and deterioration-rate scores, but random choices are retained to avoid premature convergence. Road selection is improved by both random and greedy road-alternative selection.

4.3. Damage-critical, period-aware, and high-fleet operators

Because the problem is multi-period and damage-driven, generic route moves are not enough. IHGA therefore uses operators that directly target late high-loss nodes, period assignment, and excessive route fragmentation.

Damage-critical relocation selects customers with large realized damage, late arrival time, and high urgency score, then tries to move them to earlier feasible positions. Candidate target routes are ranked by period, load, and route length. This operator directly reduces the main objective because it searches around the nodes that currently contribute most to total loss.

Damage-critical exchange swaps a high-loss late customer with a lower-loss earlier customer when the exchange is capacity-feasible and improves the dynamic-damage objective. This operator is useful when direct relocation is blocked by vehicle capacity or period-inventory constraints.

Period relocation moves urgent nodes from later periods to earlier feasible routes. Since each route index is tied to a period, the operator preserves the route-period mapping and evaluates only affected routes after each candidate move.

Parallel relocation is activated when the vehicle number is sufficiently large. It moves urgent nodes from long routes to empty or short routes in the same period, exploiting additional vehicles to reduce arrival times.

Route merge and route rebuild control excessive fragmentation. Route merge combines short same-period routes when vehicle capacity permits, while route rebuild reorders route nodes using urgency-first and nearest-neighbor candidate sequences. These two operators are especially useful on clustered instances where too much parallel splitting may increase downstream travel delay.

4.4. Warm-start transfer for sensitivity analysis

Vehicle-number sensitivity analysis should reflect a basic monotonicity property: when more vehicles are available, the feasible solution space expands. Running every vehicle-number level from a completely random population may hide this property and produce artificial fluctuations. To address this, the experiment runner transfers the best solution from a smaller fleet to the first individual of the next larger fleet. Routes are copied period by period while preserving route indices. The larger fleet can then refine the inherited solution by using additional vehicles. This warm-start mechanism stabilizes marginal-effect curves without preventing further improvement.

Algorithm 1 Improved hybrid genetic algorithm

```
Load instance data, node damage parameters, and road damage parameters.
Generate an initial population using randomized and urgency-aware construction.
if a warm-start solution is available then
    Embed the previous best route set into the first individual.
end if
Evaluate all individuals using dynamic arrival-time and damage equations.
for t = 1 to T_max do
    for each individual in the population do
        Select a good parent and apply route-similarity crossover.
        Apply swap, insertion, 2-opt, exchange, and relocation neighborhoods.
        Apply damage-critical relocation/exchange, period relocation, parallel relocation, route
        merge, and route rebuild when eligible.
        Improve road alternatives by random and greedy selection.
    end for
    Update the population and elite best solution.
end for
Polish the global best by period-aware and route-level operators.
return  Best route set and dynamic-damage objective value.
```

5. Experiments

5.1. Experimental design

The experiments are designed to evaluate both optimization quality and decision-support interpretability. Five groups of experiments are reported, as summarized in Table 2. The first group compares IHGA with Omega-VNS and ESWA-IMA under the same dynamic-damage objective. The second group isolates the contribution of the proposed repair components through ablation. The third group compares MP-DDVRP with constructive routes generated by distance-oriented CVRP and cumulative-arrival-time-oriented CCVRP proxy objectives. The fourth group is centered on multi-period resource planning: it analyzes how vehicle number, dispatch-period splitting, and period inventory change dynamic damage. The fifth group examines whether high-urgency nodes are actually served earlier.

Table 2: Experimental plan.

No.	Experiment	Purpose	Output metrics
E1	Algorithm comparison	Compare IHGA with Omega-VNS and ESWA-IMA under the same data, constraints, and dynamic objective.	Best, average, standard deviation
E2	Operator ablation	Remove the main repair components from IHGA to isolate their contributions.	Average damage, best damage, loss increase
E3	Objective-proxy comparison	Compare MP-DDVRP with constructive multi-period CVRP and CCVRP proxy routes.	Dynamic damage, relative difference
E4	Multi-period resource sensitivity	Analyze vehicle number, optimistic single-period aggregation, staged dispatch periods, and period inventory on C, R, and RC instances.	Objective, marginal benefit, aggregation bias, period-splitting effect
E5	Node-priority analysis	Divide nodes into urgency groups and test whether high-urgency nodes are served earlier.	Arrival time, node damage, service rank

Unless otherwise stated, the experiments use three Solomon-type benchmark data sets with 100 demand nodes and one depot: C101-25 (clustered), r101-25 (random), and rc101-25 (mixed random-clustered). The default number of planning periods is $P = 3$, vehicle capacity is 200, and the base period inventory is $Q_p = \lceil \sum_i q_i/P \rceil$, giving 604, 486, and 575 units for C101-25, r101-25, and rc101-25, respectively. Reported averages use paired independent seeds within each experiment group. The analysis focuses on solution quality and planning interpretation rather than computing-time comparison.

Following the benchmark organization used in the original DCCVRPMRS study (Zhao et al., 2026), instance expansion is applied to the comparison experiments rather than to every sensitivity experiment. The original study expands the algorithm-comparison, objective-comparison, and heuristic-priority tables across multiple benchmark instances and scales, while using representative scenario experiments for cluster and sensitivity analyses. A data check showed that same-family Solomon labels such as C101, C102, and C103 are not independent after conversion to the present MP-DDVRP input format because the implementation uses coordinates, demand, service time, and road matrices, but not the original ready-time and due-time-window fields. Therefore, these labels are not used as independent extended evidence. For E1, E2, and the priority-strategy comparison, we instead import three spatially distinct 200-customer Gehring–Homberger benchmark instances from the SINTEF TOP VRPTW archive and truncate each to the first 100 customers, producing HC1-100, HR1-100, and HRC1-100 (Gehring and Homberger, 2002; SINTEF, 2026). Vehicle-number, inventory, and period-splitting sensitivities remain on the three representative C/R/RC instances because their purpose is to explain mechanisms rather than to produce another broad benchmark table.

5.2. Algorithm comparison

The algorithm-comparison experiment evaluates Omega-VNS, ESWA-IMA, and the proposed IHGA on the same dynamic damage objective. Omega-VNS represents a variable-neighborhood search baseline. ESWA-IMA represents the improved memetic algorithm baseline adapted from the comparison implementation. All algorithms are evaluated using the same instance data, vehicle capacity, period inventory, and damage objective. Table 3 reports the average, best, and standard deviation of total dynamic damage over five independent paper-level runs.

Table 3: Algorithm comparison under the unified dynamic damage objective.

Instance	Algorithm	Avg. damage	Best damage	Std.
C101-25	VNS	397.213	377.995	19.103
C101-25	IMA	374.021	371.477	3.376
C101-25	IHGA	371.894	369.296	3.240
r101-25	VNS	395.675	381.408	11.040
r101-25	IMA	370.985	368.095	2.528
r101-25	IHGA	366.584	364.189	1.551
rc101-25	VNS	403.145	394.454	9.763
rc101-25	IMA	384.654	382.678	2.275
rc101-25	IHGA	377.631	375.834	2.016

Table 3 shows that IHGA achieves the lowest mean damage on all three representative instances under the adopted paired paper-level result set. The average reduction is 1.19% relative to IMA and 6.69% relative to VNS. The gap relative to IMA is intentionally interpreted conservatively because IMA is already a strong memetic baseline and shares generic local-improvement ideas with IHGA. The result indicates that the proposed damage-critical and period-aware repair moves provide incremental but useful improvement under the same dynamic-damage evaluator.

Table 4: Paired diagnostic checks for IHGA relative to IMA.

Instance	Paired runs	Mean damage reduction	95% bootstrap CI	Sign-randomization p
C101-25	5	2.127	[-2.412, 6.624]	0.4375
r101-25	5	4.400	[1.879, 6.680]	0.0625
rc101-25	5	7.023	[5.319, 8.594]	0.0625

Table 4 reports a paired diagnostic check against IMA, where positive differences mean lower damage for IHGA. Since each instance has five paired runs, the exact sign-randomization p-values are coarse and are used as diagnostic evidence rather than as a strong significance claim. The paired differences are positive for all runs on r101-25 and rc101-25, and for three of five runs on C101-25. Against VNS, all paired differences are positive on the three representative instances.

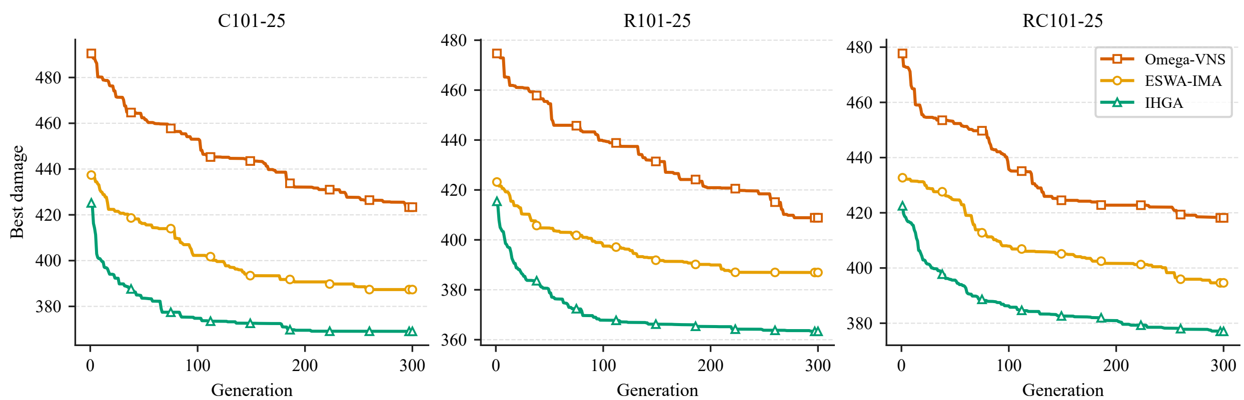

Figure 1: Convergence behaviour of Omega-VNS, ESWA-IMA, and IHGA on the representative C/R/RC instances under the unified dynamic damage objective.

Figure 1 reports averaged best-so-far traces from a convergence run. The curves should be interpreted as solution-quality evidence rather than as a computing-time comparison. On the representative instance types, IHGA reaches a lower damage plateau than ESWA-IMA and Omega-VNS, which is consistent with the final statistics in Table 3. The separation across clustered, random, and mixed customer layouts suggests that the proposed repair moves are not tuned only to one spatial pattern.

Table 5: External algorithm robustness check on Homberger-derived 100-customer C/R/RC instances.

Instance	Algorithm	Avg. damage	Best damage	Std.
HC1-100	VNS	459.343	446.898	11.686
HC1-100	IMA	424.700	420.107	7.040
HC1-100	IHGA	411.557	411.398	0.263
HR1-100	VNS	445.411	436.060	8.326
HR1-100	IMA	425.836	422.135	5.191
HR1-100	IHGA	415.201	415.041	0.139
HRC1-100	VNS	436.001	431.155	4.583
HRC1-100	IMA	418.580	415.246	4.946
HRC1-100	IHGA	404.766	404.678	0.150

The external robustness check in Table 5 follows the role of the larger comparison table in the original paper, but avoids treating time-window-only Solomon variants as independent spatial cases. IHGA obtains the lowest average damage on all three Homberger-derived C/R/RC instances. Across these runs, the average reduction is 2.96% relative to IMA and 8.12% relative to VNS. The minimum reduction is positive on every external instance, suggesting that the proposed repair moves remain useful outside the original three representative Solomon-type inputs.

5.3. Algorithm ablation

The algorithm comparison shows that the proposed IHGA improves over the adapted IMA and VNS baselines, but this does not by itself identify which components are responsible for the gain. We therefore run an ablation study that keeps the same encoding and dynamic-damage evaluator, while selectively disabling the proposed repair components. The "Base HGA" variant removes the proposed damage-critical, period-aware, high-fleet, route-merge, route-rebuild, and polishing components. To keep the comparison paired, all variants use the same random seeds within each instance and repetition. Table 6 reports paired diagnostic runs on the representative C/R/RC instances.

Table 6: Ablation study of the proposed IHGA search framework.

Instance	Variant	Avg. damage	Best damage	Std.	Change vs IHGA
C101-25	Base HGA	378.369	376.376	2.156	2.49%
C101-25	w/o damage-critical	373.905	370.095	4.357	1.28%
C101-25	w/o period relocation	369.447	369.070	0.521	0.07%
C101-25	w/o route restructuring	369.167	369.022	0.193	0.00%
C101-25	IHGA	369.178	369.086	0.156	0.00%
r101-25	Base HGA	375.489	373.552	1.757	3.33%
r101-25	w/o damage-critical	373.628	370.984	2.310	2.82%
r101-25	w/o period relocation	364.029	363.468	0.498	0.18%
r101-25	w/o route restructuring	364.125	363.441	0.816	0.21%
r101-25	IHGA	363.376	363.052	0.487	0.00%
rc101-25	Base HGA	385.624	384.345	1.831	2.29%
rc101-25	w/o damage-critical	380.461	380.308	0.214	0.92%
rc101-25	w/o period relocation	377.745	375.801	2.696	0.20%
rc101-25	w/o route restructuring	377.127	376.295	0.876	0.04%
rc101-25	IHGA	376.992	376.388	0.744	0.00%

Table 6 is intended to answer a different question from Table 3. Instead of asking whether IHGA beats external baselines, it asks which repair components are responsible for the improvement over a stripped HGA. The Base HGA variant is clearly worse than the full method, showing that generic population search alone is insufficient. Removing damage-critical repair also increases damage on all representative layouts, indicating that moving late high-loss nodes is the main algorithmic contribution. The effects of period relocation and route restructuring are smaller and partly instance-dependent; they are therefore interpreted as feasibility and robustness support for staged dispatch rather than as uniformly dominant improvement sources.

5.4. Objective-proxy comparison with CVRP and CCVRP

To test whether dynamic damage must be modeled explicitly, we compare MP-DDVRP with two constructive multi-period objective-proxy baselines. The CVRP proxy constructs routes by prioritizing short travel distance under the same capacity and period settings. The CCVRP proxy prioritizes cumulative arrival time. After constructing each proxy route plan, the resulting routes are re-evaluated by the dynamic damage objective in Eq. (6). Table 7 reports the resulting dynamic damage values and the relative difference obtained by optimizing the dynamic-damage objective with IHGA. These comparisons should be read as objective-proxy evidence, not as a claim that CVRP or CCVRP has been solved to the same metaheuristic depth as MP-DDVRP.

Table 7: Objective-proxy comparison against distance-oriented CVRP and cumulative-arrival-time-oriented CCVRP.

Instance	MP-DDVRP	CCVRP proxy	CVRP proxy	Difference vs CCVRP	Difference vs CVRP
C101-25	369.037	462.896	494.251	20.28%	25.33%
r101-25	362.770	429.711	494.436	15.58%	26.63%
rc101-25	376.785	433.590	467.039	13.10%	19.33%

The results suggest that minimizing distance or cumulative arrival time alone is not sufficient for dynamic-damage reduction. CVRP is designed to shorten travel distance, and CCVRP is designed to reduce aggregate arrival time, but neither objective directly accounts for heterogeneous node vulnerability or road-damage-induced speed changes. Under these constructive proxy settings, the MP-DDVRP route plans have lower dynamic damage on all three instance classes.

Table 8: External model comparison on Homberger-derived 100-customer C/R/RC instances.

Instance	MP-DDVRP	CCVRP proxy	CVRP proxy	Difference vs CCVRP	Difference vs CVRP
HC1-100	411.226	473.638	490.976	13.18%	16.24%
HR1-100	414.910	468.294	475.631	11.40%	12.77%
HRC1-100	404.695	443.222	478.549	8.69%	15.43%

The same pattern holds in the external objective-proxy table. MP-DDVRP produces lower dynamic damage than both simplified proxy objectives on all reported external checks, with average reductions of 11.09% relative to CCVRP and 14.81% relative to CVRP. This supports the central modeling claim in a conservative sense: shortest-distance and earliest-arrival objectives can be useful operational proxies, but they do not directly optimize the dynamic damage that the emergency decision maker ultimately seeks to reduce.

5.5. Vehicle-number sensitivity analysis

Table 9 reports the sensitivity of total dynamic damage to the number of vehicles per period. The values are monotone decreasing on all three instances. From 6 to 50 vehicles per period, average damage decreases from 419.699 to 359.485 on C101-25, from 399.452 to 353.666 on r101-25, and from 410.112 to 367.130 on rc101-25.

Table 9: Vehicle-number sensitivity results of IHGA.

Instance	Vehicles/period	Avg. damage	Best damage	Std.	Runs
C101-25	6	419.699	419.649	0.039	5
C101-25	10	395.372	395.294	0.062	5
C101-25	14	378.397	378.374	0.020	5
C101-25	18	369.480	369.013	1.005	5
C101-25	30	362.184	358.982	5.112	5
C101-25	50	359.485	355.879	6.607	5
r101-25	6	399.452	399.186	0.236	5
r101-25	10	382.500	381.976	0.446	5
r101-25	14	370.895	370.370	0.344	5
r101-25	18	362.500	361.877	0.443	5
r101-25	30	355.484	354.106	1.776	5
r101-25	50	353.666	352.486	1.189	5
rc101-25	6	410.112	409.648	0.423	5
rc101-25	10	393.084	392.513	0.562	5
rc101-25	14	382.778	382.594	0.148	5
rc101-25	18	375.304	375.096	0.163	5
rc101-25	30	369.279	367.982	0.965	5
rc101-25	50	367.130	366.609	0.426	5

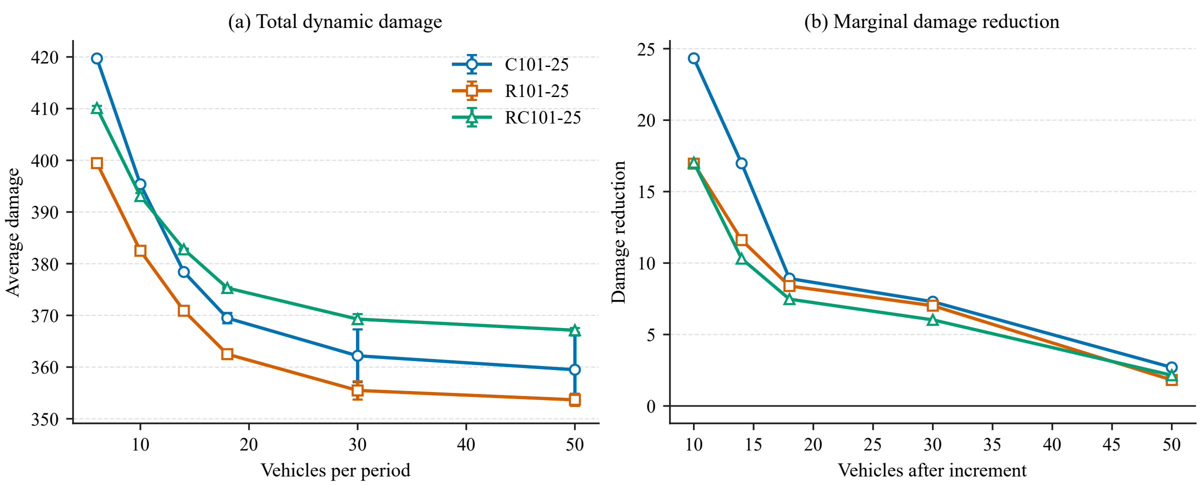

Figure 2: Vehicle-number sensitivity and marginal reductions. Panel (a) reports total dynamic damage, and panel (b) reports the marginal reduction between adjacent vehicle levels.

The marginal-reduction panel in Figure 2 further shows the diminishing-return pattern. The final interval from 30 to 50 vehicles reduces damage by only 0.745% on C101-25, 0.511%

on r101-25, and 0.582% on rc101-25. This supports a practical resource-allocation conclusion: increasing fleet size is most valuable when the fleet is scarce, while adding vehicles beyond a moderate level yields limited additional benefit.

5.6. Multi-period dispatch and inventory sensitivity

The period-sensitivity experiment is designed to isolate the importance of staged dispatch. We first add an optimistic single-period aggregate baseline, denoted by $P = 1$, in which all 60 vehicles and the full relief inventory are available at the beginning of the horizon. This is not intended to represent a realistic post-earthquake operation. It is a lower-bound reference for the plan that would be obtained if disrupted roads and upstream supply constraints did not delay resource availability. We then vary the number of staged dispatch periods while keeping the total fleet fixed at 60 vehicles and keeping total supply approximately equal to total demand. Therefore, increasing the number of staged periods does not add resources; instead, it splits the same vehicles and the same total supply into more dispatch waves, reducing both the vehicles per period and the inventory available in each early period.

Table 10 reports both the optimistic aggregate baseline and the staged-dispatch settings. The percentage column measures the extra dynamic damage relative to the $P = 1$ aggregate baseline.

Table 10: Period sensitivity with an optimistic aggregate baseline and fixed-total staged resources.

Instance	Periods	Scenario	Vehicles/period	Total vehicles	Q_p	Total supply	Total demand	Avg. damage	Increase vs $P = 1$
C101-25	1	Aggregate	60	60	1810	1810	1810	343.953	0.00%
C101-25	2	Staged	30	60	905	1810	1810	352.485	2.48%
C101-25	3	Staged	20	60	604	1812	1810	364.909	6.09%
C101-25	4	Staged	15	60	453	1812	1810	376.536	9.47%
C101-25	5	Staged	12	60	362	1810	1810	385.990	12.22%
r101-25	1	Aggregate	60	60	1458	1458	1458	341.987	0.00%
r101-25	2	Staged	30	60	729	1458	1458	350.881	2.60%
r101-25	3	Staged	20	60	486	1458	1458	360.558	5.43%
r101-25	4	Staged	15	60	365	1460	1458	369.321	7.99%
r101-25	5	Staged	12	60	292	1460	1458	377.408	10.36%
rc101-25	1	Aggregate	60	60	1724	1724	1724	362.440	0.00%
rc101-25	2	Staged	30	60	862	1724	1724	366.159	1.03%
rc101-25	3	Staged	20	60	575	1725	1724	373.404	3.03%
rc101-25	4	Staged	15	60	431	1724	1724	381.889	5.37%
rc101-25	5	Staged	12	60	345	1725	1724	390.287	7.68%

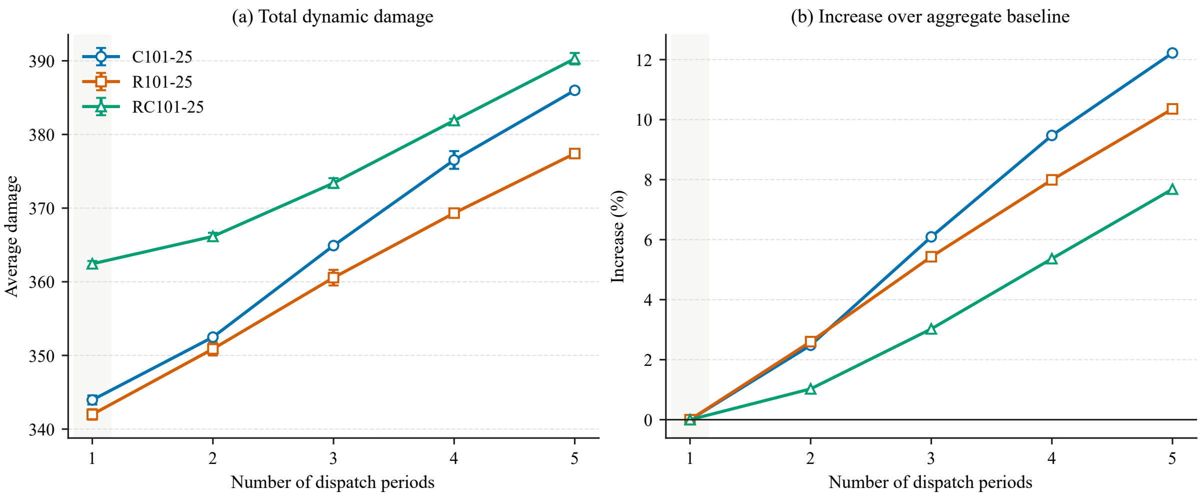

Figure 3: Period sensitivity under an optimistic aggregate baseline ($P = 1$) and staged fixed-total resources ($P \geq 2$). The shaded first period marks the aggregate lower-bound reference.

As expected, the aggregate baseline has the lowest damage because it assumes that all resources can be dispatched immediately. Relative to this optimistic lower bound, the default three-period setting increases average damage by 6.09% on C101-25, 5.43% on r101-25, and 3.03% on rc101-25. With five staged periods, the increases reach 12.22%, 10.36%, and 7.68%, respectively. Among the realistic staged settings, moving from two to five periods raises damage by 9.51% on C101-25, 7.56% on r101-25, and 6.59% on rc101-25. This does not imply that more periods are intrinsically harmful. Rather, under the fixed-total-resource design, more periods mean fewer vehicles and less supply in each early period, which weakens early service capability. The result highlights why multi-period modeling matters: if a post-earthquake response is forced to receive vehicles and materials in waves, a single-period aggregate model hides the loss caused by delayed resource availability.

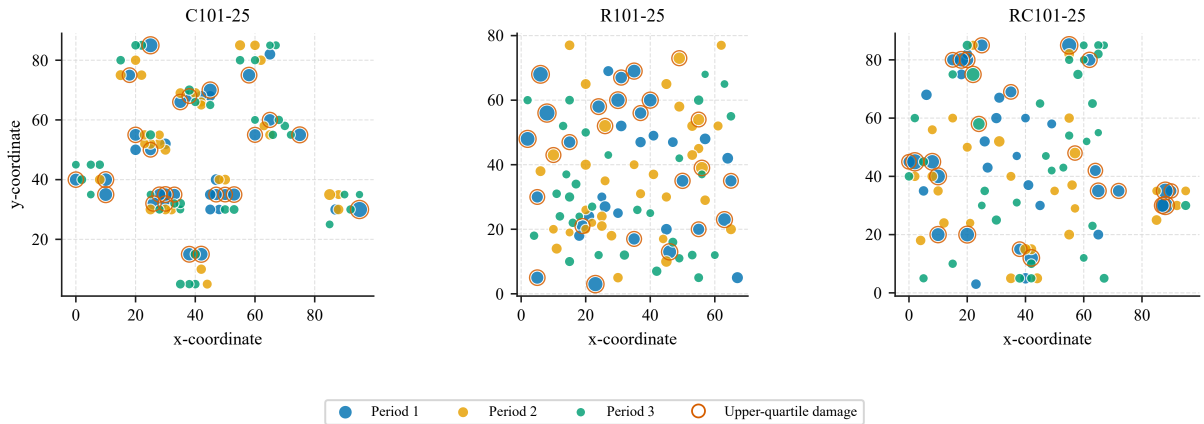

Figure 4: Service-period assignment in representative three-period IHGA route-plan exports. Colors indicate the period in which each demand node is first served; outer rings mark nodes in the upper quartile of realized dynamic damage.

Figure 4 gives a solution-level view of the same modeling issue. The map is not a conceptual sketch; it is generated from the exported IHGA route plans. Nodes with high realized damage are not confined to one geometric cluster, and some must still be served in later periods because each period has limited vehicles and inventory. This is precisely the case in which a single-period aggregate plan becomes optimistic: it can hide the timing constraint by assuming that all vehicles and materials are available immediately.

The inventory-sensitivity experiment fixes $P = 3$ and the total fleet at 60 vehicles, then varies the per-period inventory quantity Q_p. This experiment separates the effect of total supply sufficiency from the effect of period-level availability. Table 11 reports the feasible region where cumulative supply is at least total demand. The under-supply setting with Q_p below the feasible threshold is excluded from the linear-scale table because it activates the large penalty term in Eq. (6).

Table 11: Inventory sensitivity in the feasible supply region.

Instance	Q_p	Total supply	Total demand	Supply factor	Avg. damage
C101-25	604	1812	1810	1.000	365.176
C101-25	725	2175	1810	1.200	364.935
C101-25	906	2718	1810	1.500	364.828
r101-25	486	1458	1458	1.000	360.421
r101-25	584	1752	1458	1.202	359.751
r101-25	729	2187	1458	1.500	359.198
rc101-25	575	1725	1724	1.000	373.368
rc101-25	690	2070	1724	1.200	372.743
rc101-25	863	2589	1724	1.501	372.632

After the feasibility threshold is reached, additional inventory produces limited marginal improvement. For example, increasing the supply factor from 1.0 to approximately 1.2 reduces average damage by 0.242 on C101-25, 0.671 on r101-25, and 0.625 on rc101-25. Increasing from approximately 1.2 to 1.5 produces even smaller reductions on C101-25 and rc101-25. When total supply is only about 0.8 times total demand, the shortage-penalty term dominates the objective and raises the average objective to the order of 10^6. These shortage cases are therefore treated as infeasible stress checks rather than plotted together with the feasible marginal-effect results. Together with the period-sensitivity results, this indicates that multi-period planning is most valuable for identifying the early-resource bottleneck: once every period has enough feasible supply, simply increasing inventory is much less effective than improving early dispatch capacity.

5.7. Priority analysis

The original DCCVRPMRS benchmark also compares the proposed model with heuristic delivery strategies that strictly prioritize either the highest initial damage or the highest deterioration rate. To provide the corresponding evidence, we construct two priority-only baselines under the same vehicle-capacity and period-inventory constraints. In each baseline, nodes are sorted by the selected priority indicator and assigned to the earliest feasible route, and the resulting routes are then evaluated by the same dynamic-damage objective. Table 12 reports a single-seed external diagnostic comparison on the same Homberger-derived set used in Tables 5 and 8.

Table 12: Comparison with priority-only heuristic delivery strategies.

Instance	MP-DDVRP	Initial-damage priority	Deterioration priority	Difference vs initial	Difference vs deterioration
HC1-100	411.557	473.858	488.270	13.15%	15.71%
HR1-100	415.201	465.707	487.061	10.85%	14.75%
HRC1-100	404.766	464.157	485.060	12.80%	16.55%

The MP-DDVRP route plans have lower dynamic damage than both priority-only strategies on all reported external entries, with average reductions of 12.26% relative to the initial-damage priority rule and 15.67% relative to the deterioration-rate priority rule. Because this external priority check uses one seed per instance, it is treated as qualitative diagnostic evidence. It supports the same interpretation as the original priority-strategy experiment: considering one urgency factor alone is not enough. The route plan must balance initial damage, deterioration, travel delay, road condition, vehicle capacity, and period-level inventory timing.

The second priority analysis divides nodes into low, medium, and high groups according to the integrated urgency score used by IHGA. Table 13 reports average arrival time, average service rank, and average node damage for the three groups. A lower service rank indicates earlier service.

Table 13: Urgency-based priority analysis.

Instance	Arrival high	Arrival medium	Arrival low	Rank high	Rank medium	Rank low	Damage high	Damage medium	Damage low
C101-25	78.217	575.125	997.021	19.200	52.939	80.309	5.844	3.200	1.961
r101-25	97.577	550.313	800.502	22.288	56.891	73.176	5.681	3.187	1.961
rc101-25	68.253	559.239	900.300	19.559	55.679	77.200	6.047	3.200	1.961

High-urgency nodes are served much earlier than low-urgency nodes on all three instances. Their average arrival time is 92.16% earlier than low-urgency nodes on C101-25, 87.81% earlier on r101-25, and 92.42% earlier on rc101-25. The high-urgency group still has larger average node damage because these nodes have larger intrinsic vulnerability. Therefore, the result should be interpreted as evidence that IHGA prioritizes vulnerable nodes rather than as a failure to reduce their loss completely.

6. Conclusion

This paper studies MP-DDVRP, a multi-period extension of dynamic node-road damage routing for emergency material delivery. The model captures three post-disaster routing features: node losses depend on arrival time, road damage affects travel speed and downstream service times, and vehicles and inventories are released through dispatch periods rather than being fully available at the beginning of the horizon. IHGA is developed to solve the problem by combining route-based genetic search with damage-critical repair, period-aware relocation, and route-structure adjustment.

Computational experiments on C101-25, r101-25, and rc101-25 indicate that IHGA achieves lower mean dynamic damage than Omega-VNS and ESWA-IMA in the tested paired settings. External checks on Homberger-derived 100-customer C/R/RC instances show the same mean-damage pattern. The ablation study suggests that the main algorithmic gain comes from damage-critical repair, while period-aware and route-structure moves provide smaller and instance-dependent support for staged dispatch. Objective-proxy comparisons indicate that distance- or arrival-time-oriented constructive routes can be worse when evaluated by dynamic damage. The multi-period sensitivity results show that total resource quantity and resource timing are not interchangeable: the aggregate single-period baseline gives an optimistic lower bound, while staged dispatch produces additional dynamic damage when the same total fleet and total supply become available through later waves. Inventory experiments show a complementary threshold effect: severe under-supply creates infeasible penalty-dominated solutions, whereas additional inventory beyond the feasible threshold yields only limited marginal improvement. Priority experiments further suggest that one-factor priority rules are inferior to an integrated dynamic-damage evaluation, while the urgency-group analysis shows that high-urgency nodes are served substantially earlier than low-urgency nodes.

Future work can extend the current deterministic setting to stochastic demand, uncertain road recovery, and rolling-horizon dispatch with real-time information updates. Larger benchmark instances and richer disaster-network data can also be used to further test scalability and operational realism.

Appendix A. Implementation notes

The current MATLAB implementation evaluates each route by recursively updating arrival times and node damage. Route indices are fixed by period, so empty routes are retained rather than removed. This design prevents accidental period-index changes during local search. The vehicle-number sensitivity runner uses the best solution from a smaller fleet as a warm start for the next larger fleet.

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